Week 4 – Order Management System

1. Orders & Order_Details Table Design

The Orders table stores order information such as Order_ID, Customer_ID, Order_Date, Total_Amount and Order_Status.

The Order_Details table stores Order_Detail_ID, Order_ID, Product_ID, Quantity and Price.

Primary keys uniquely identify records, while foreign keys establish relationships between Customer, Orders, Product and Order_Details. NOT NULL, UNIQUE and CHECK constraints maintain data accuracy.

2. Updated ER Diagram

CUSTOMER → ORDERS → ORDER_DETAILS ← PRODUCT

- Customer to Orders: One-to-Many
- Orders to Order_Details: One-to-Many
- Product to Order_Details: One-to-Many

3. Sample Customer, Order & Order_Details Data

Sample customers include Arun Kumar, Priya Devi and Karthik Raj.

Sample products include Keyboard, Mouse, Monitor, USB Cable and Headphones.

Order details contain the products purchased, quantity, price and total amount for each order.

4. Customer Order History & Purchase Analysis Report

SQL queries are used to display customer order history, product-wise orders, the number of times each product was ordered, total quantity sold, customers with the maximum number of orders, total spending by each customer, average order value and total sales amount.

5. Query Output Screenshots

Screenshots from MySQL Workbench should be added for the following queries:

1. Complete Order Information
2. Customer Order History
3. Product-wise Order Report
4. Number of Times Each Product Was Ordered
5. Total Quantity of Products Sold
6. Customers with Maximum Number of Orders

7. Total Spending by Each Customer
8. Average Order Value
9. Total Sales Amount